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**Mitochondrial Phylogeography and Species Delimitation of Sexual and Asexual
populations of the Ostracod *Cypridopsis vidua* Across Continents**

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Abstract

This study investigates the evolutionary history, genetic diversity, and reproductive biology of the widespread non-marine ostracod *Cypridopsis vidua*. We curated and analysed 170 mitochondrial COI sequences from specimens collected across five regions, Belgium, North and South America, Turkey and New Caledonia. Phylogenies under the best-fit substitution model were reconstructed with both maximum-likelihood and Bayesian inference. Four statistical species-delimitation methods (ABGD, ASAP, bPTP, KoT) were applied to identify genetic species. Our results reveal remarkably shallow COI divergence and a star-like haplotype network with extensive haplotype sharing, indicating a single, genetically cohesive lineage including both sexual and asexual populations. Nevertheless, sensitive delimitation methods identify multiple genetic species, suggesting the presence of incipient cryptic diversity that lacks strict geographic segregation and warrants morphological validation. The first molecular analyses of the recently discovered North American sexual population, a rarely reported phenomena for *C. vidua*, highlights mixed reproduction as a key driver for mitochondrial cohesion, contrasting patterns seen in parthenogenetic ostracods. Finally, we discuss the limitations of COI barcoding for resolving fine-scale genetic divergence and advocate a multi-locus, integrative taxonomic framework to refine species boundaries and elucidate the role of reproductive transitions in shaping ostracod diversity.

Keywords: *Cypridopsis vidua*; Ostracoda; COI barcoding; Cryptic diversity; Parthenogenesis – Sexual reproduction; Phylogenetic inference; Species delimitation.

1. Introduction

Ostracods, small crustaceans within the class Ostracoda of the subphylum Crustacea, represent one of nature's most enduring evolutionary success stories. Encased in bivalved, calcified carapaces, these organisms, typically ranging from 0.2 to a few millimetres in length, have thrived for over 500 million years, since the Cambrian period, surviving mass extinctions that obliterated countless other taxa (Maddocks, 1982; Oakley *et al.*, 2013). Their fossilised shells, preserved in sedimentary layers across the globe, constitute the most extensive fossil record of all living arthropods, offering a window into deep evolutionary time (Wolfe *et al.*, 2016). Ostracods inhabit an extraordinary range of environments, from the abyssal depths of oceans to freshwater lakes, rivers, caves, and even interstitial groundwater systems (Smith *et al.*, 2015). This thesis centres on *Cypridopsis vidua*, a non-marine ostracod species long considered exclusively parthenogenetic, reproducing asexually through unfertilised eggs. However, recent discoveries of sexual populations (Martens *et al.*, 2023; Zhai *et al.* 2023; Rossetti *et al.*, 2024), have prompted a re-examination of its reproductive strategies, genetic diversity, and evolutionary history.

The discovery of sexual reproduction in *C. vidua* challenges its traditional classification and raises intriguing questions about cryptic diversity, genetically distinct yet morphologically similar lineages often hidden within a single species (Bickford *et al.*, 2007). Until recently, *C. vidua* was considered an obligate parthenogen, with the first unequivocal observation of sexual males and females reported by Martens *et al.* (2023) from a population in Woods Hole, Massachusetts, USA. This late discovery can be attributed to the extreme difficulty in identifying males without dissection, as the testicular tubes and Zenker organ are barely visible through the carapaces in transparency. The authors only became aware that the population contained males when discovering abundant sperm during dissection of a female specimen. Additionally, males are smaller than females with less pronounced external ornamentation (see **Figure 1**), and in mixed populations, the ratio of males to females may be quite low. These elusive characteristics meant that males could have been easily overlooked in populations that were generally assumed to be all-female, despite *C. vidua* being one of the most studied non-marine ostracod species.

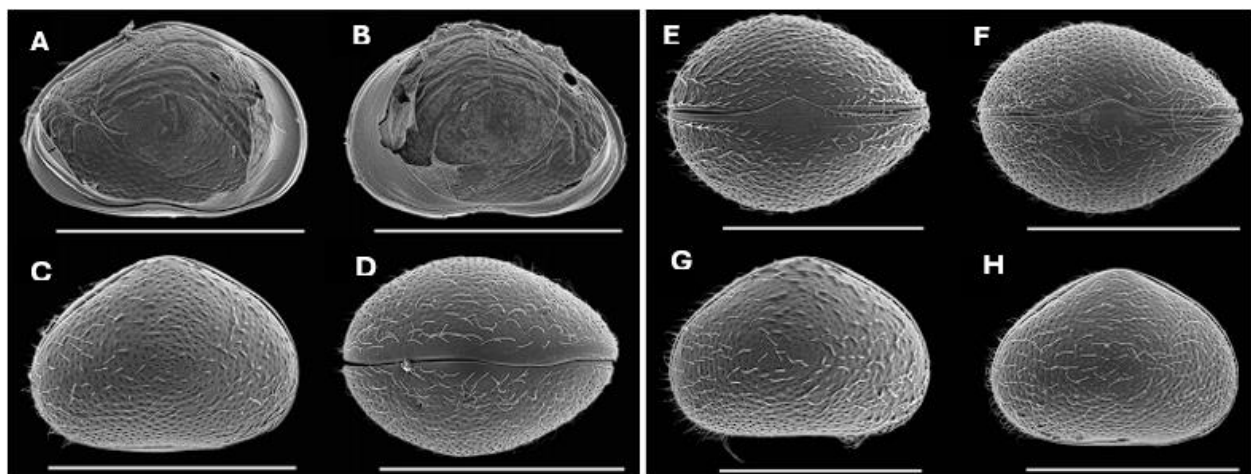


Figure 1. *Cypridopsis vidua*, ♂ (A, B, C, D) and *Cypridopsis vidua*, ♀ (E, F, G, H). (Martens *et al.*, 2023)

Reproductive strategies in non-marine ostracods are notably varied, with different species exhibiting distinct reproductive modes ranging from obligate parthenogenesis to sexual reproduction, but these strategies are generally fixed. Sexual reproduction, the ancestral mode, involves male and female gametes fusing to produce genetically diverse offspring, as seen in many candonids and cytherids (Martens, 1998). Parthenogenesis, where females produce offspring without fertilization of the egg, enables rapid population growth and colonization of isolated habitats, offering short-term evolutionary advantages (Bell, 1982). Many species of the, Cyprididae like *C. vidua*, including for example the best studied species *Eucypris virens* and *Heterocypris incongruens*, exhibit mixed reproduction, a mixture of sexual and asexual reproduction in the same population to balance genetic variability with reproductive efficiency (Martens, 1998). Geographical parthenogenesis, where asexual populations are wider distributed than sexual ones, reveal geographic separation of asexual and sexual populations, as observed in species like *Eucypris virens* (Schmit *et al.*, 2013). Additionally, putative ancient asexuals, such as darwinulid ostracods, reproduce exclusively parthenogenetic and defy expectations by persisting without sex for millions of years (Martens *et al.*, 2003; Schön *et al.*, 2009).



Figure 2. *Cypridopsis vidua*, copula of male (right) and sexual female (left) (no scale). (Martens *et al.* 2023)

Historically, *Cypridopsis vidua* was considered as a widespread, obligate parthenogen, incapable of sexual reproduction (Havel & Hebert, 1989). However, recent discoveries have fundamentally challenged this classification (see **Figure 2**). Martens *et al.* (2023) documented the first known sexual population of *C. vidua* containing functional males alongside both sexual and asexual females, collected from a freshwater pond in Woods Hole, Massachusetts, USA. Notably, they observed copulation events and direct evidence of insemination, providing unambiguous proof that males are reproductively active. This revelation signifies that *C. vidua* exhibits mixed reproduction, a strategy that combines the evolutionary advantages of both sexual and asexual reproductive modes. Following Martens *et al.* (2023)'s initial discovery in the United States, additional sexual populations have been documented from Southwest China by Zhai *et al.* (2023), representing the first records of males outside the USA, and from northern Italy by Rossetti *et al.* (2024), marking the first record of males in Europe. These discoveries suggest that sexual

populations of *C. vidua* are likely much more widespread than previously known, as the recognition of males requires dissection of individuals and can easily be overlooked.

The finding of sexual reproduction in *C. vidua* offers a critical explanatory framework for previous molecular observations. Earlier genetic studies had already noted unusually high levels of allozyme and mitochondrial cytochrome c oxidase subunit I (COI) diversity within nominally asexual populations (Cywinska & Hebert, 2002). At the time, these patterns were attributed to processes such as recurrent colonisation or in situ mutational accumulation. However, the existence of functional males and sexual reproduction suggests that recombination events, although rare or localised, could have contributed substantially to maintaining this genetic variability over time. Furthermore, ongoing molecular work aims to systematically resolve the genetic architecture of *C. vidua*, with the anticipation of discovering multiple cryptic species within what has historically been treated as a single morphospecies.

The revelation of sexual *C. vidua* populations underscores the need for molecular approaches to uncover cryptic diversity, as demonstrated in studies of *Eucypris virens* (over 35 cryptic species identified; Bode *et al.*, 2010) and *Heterocypris salina* (Kilikowska *et al.*, 2024). Similar integrative analyses of Australian *Bennelongia* species have revealed previously unrecognised taxa with and without cryptic diversity, emphasising the power of combining molecular phylogenetics with morphological data (Martens *et al.*, 2013, 2015) for integrative taxonomy. Phylogenetic analyses further illuminate ostracod evolution. Broad-scale studies, like Oakley *et al.* (2013), used phylotranscriptomics and fossils to confirm Ostracoda monophyly and date its origins, while finer-scale work on ancient lake endemics, like Lake Baikal's candonids, links cryptic speciation to geological history (Karanovic & Sitnikova, 2017a, 2017b; Schön & Martens, 2012). For *C. vidua*, the current study provides data on reproductive mode transitions and genetic which may inform future research on ostracod adaptability and evolutionary trajectories, thereby contributing to broader understandings of ostracod biology and biodiversity.

Beyond advancing knowledge of ostracod reproductive modes, the discovery of sexuality in *C. vidua* has broader evolutionary implications. It challenges the long-standing view that obligate asexuality represents an evolutionary cul-de-sac characterised by reduced genetic variability and limited adaptability. Instead, it supports emerging hypotheses suggesting that rare sexual events or mixed reproductive strategies can sustain genetic diversity, prevent mutational meltdown, and enable long-term persistence even in lineages predominantly reproducing asexually (Schön *et al.*, 2009). This aligns with patterns observed in other ostracods, such as *Eucypris virens*, where mixed reproductive strategies underpin extensive cryptic speciation and ecological success (Bode *et al.*, 2010).

Methodological advances have further enhanced the capacity to detect and characterise sexual structures in small-bodied crustaceans. Martens *et al.* (2023) employed polychromatic polarisation microscopy, a novel imaging technique that highlights weakly birefringent structures in vivid colours without staining, to reveal subtle soft-part morphologies, including the hemipenes and prehensile palps of male *C. vidua*. This approach allows for a non-invasive, objective examination of minute anatomical features, offering new opportunities to revisit historical material and

potentially uncover previously overlooked sexual populations in this and other presumed asexual species.

1.1. Objectives

A key objective of this study is to test for the phylogenetic position of sexual populations of *C. vidua* and to identify the factors shaping possible phylogeographic patterns through detailed haplotype network and phylogenetic tree analyses. To accomplish this, the study analysed more than 100 specimens across the species' range, and the application of advanced statistical techniques to reveal non-random geographic distribution patterns and hidden genetic structures.

In parallel, the study examines cryptic genetic diversity within *C. vidua*, employing high-resolution statistical methods for species delimitations to assess whether sexual and asexual populations represent distinct evolutionary lineages or merely variations within a single species. This effort builds on previous research that uncovered cryptic species in other ostracods and involved sequencing both mitochondrial and nuclear markers. The resulting data were used to reconstruct detailed phylogenies, clarifying whether historical classifications have masked underlying genetic divergence and whether there are cryptic speciation events that demand a taxonomic revision.

By integrating molecular dating with distribution data, the study aims to trace the evolutionary history and geographic distribution of sexual *C. vidua* and evaluate their adaptive implications. These results lay the groundwork for future investigations into the evolutionary history and adaptive significance of mixed reproductive modes.

Guiding this study are three critical questions that will contribute to a broader understanding of reproductive strategies in facultatively asexual organisms:

- **First: how do geographic distributions differentiate sexual and asexual *C. vidua* populations?** Detailed mapping and analysis of haplotype networks across diverse locations will allow us to discern if sexual populations are isolated or intermixed with asexual ones, this approach enables us to assess patterns of dispersal and historical demographic events that might have contributed to the current arrangement of reproductive types within the species.
- **Second: does molecular evidence support the presence of cryptic species, thereby challenging traditional taxonomic classifications?** The aim is to determine whether the observed genetic variation exceeds the levels typically found within a single morphospecies. If the genetic divergence is significant according to specialized statistical methods for genetic species delimitations, it would indicate the presence of additional, possibly cryptic species that have not been detected by conventional morphological analyses.

- And finally, **what do shifts in reproductive modes reveal about the species' evolutionary trajectory and the maintenance of sexual reproduction in a predominantly asexual context?** One of the longstanding evolutionary puzzles is why sexual reproduction persists in species where asexuality appears to offer immediate benefits, such as higher reproductive rates. Through haplotype network analyses, this thesis provides preliminary insights into the genetic relationships among sexual and asexual *C. vidua* lineages, offering a foundation for understanding the persistence of sexual reproduction.

In summary, this study seeks to provide new insights into how sexual reproduction persists within a lineage that predominantly reproduces asexually, and to refine our understanding of species boundaries within this taxonomic morphospecies. This will be achieved by providing an understanding of these shifts by addressing the aforementioned two interrelated questions: the spatial distribution of reproductive types, the presence and extent of cryptic genetic diversity, and by combining detailed geographic mapping, comprehensive molecular analyses, and robust species delimitation methods.

2. Materials and Methods

2.1. Molecular Analyses

Sexual specimens of *Cypridopsis vidua* from Italy were provided by Professor Giampaolo Rossetti; all other sequences from asexual and sexual specimens, including those from Belgian SPEEDY sites, North and South America, Ecuador, New Caledonia, Turkey, and Brazil, had been previously generated, organized, and quality-checked by two research assistants at the RBINS, I. Schön and an intern from the Interuniversity Master of Science in Marine and Lacustrine Science and Management (Oceans and Lakes program), so no additional sequencing was required for these samples. All specimens destined for DNA extraction were initially preserved in 100% ethanol to maintain tissue integrity and prevent degradation. Prior to DNA extraction, each specimen was carefully removed from ethanol storage and briefly rinsed with distilled water. After rinsing, specimens were transferred to DNA-free 1.5 mL microcentrifuge tubes containing 180 µL of Buffer ATL (lysis buffer) and 20 µL of Proteinase K as provided in the Qiagen DNeasy Blood & Tissue kit, following a modified protocol of Schön *et al.* (2012b). Tubes were incubated overnight at 56 °C in a ThermoMixer. This prolonged incubation ensured complete digestion of exoskeletal and soft tissues, facilitating maximum release of genomic DNA from cellular compartments. Each sample's concentration and purity were measured using a NanoDrop 2000 spectrophotometer, checking the absorbance at 260 nm and 280 nm. Samples with a 260/280 ratio between 1.8 and 4.0 were considered clean enough for the next steps. All DNA extracts were stored at –20 °C.

PCR reactions were carried out in a T Personal Thermocycler (Biometra). A part of the mitochondrial COI gene was amplified, using a modification of the universal invertebrate primers (Folmer *et al.*, 1994). The forward primer sequence (5'-GGTCAACAAATCATAAAGATATTGG-3') and the reverse primer sequence (5'-TAAACTTCAGGGTGACCAAAAAATCA-3') amplify a fragment of about 658 bp, which is widely used for DNA barcoding in many invertebrates. Each 25 µL reaction contained 12.5 µL of HotStarTaq Master Mix (1.5–2 mM MgCl₂, 200 µM dNTPs, Tris·Cl, KCl, (NH₄)₂SO₄, 1.25 U Taq polymerase), along with 1 µM of each primer and 1 µL of genomic DNA. Nuclease-free water was added to reach 25 µL. COI amplification was done using the following program: an initial 15-minute step at 95 °C to activate the HotStarTaq polymerase, followed by 40 cycles of 95 °C for 50 seconds (denaturation), 46 °C for 50 seconds (annealing), and 72 °C for one minute (extension). After those cycles, a final extension at 72 °C for 10 minutes was done. The 46 °C annealing temperature was chosen based on the primers' melting temperatures and previously successful protocols. Once PCR was finished, the success of amplifications was verified using agarose gel electrophoresis and subsequent staining with Midori Green. The gel was run at 100 V for 1 hour 20 minutes on a Bio-Rad PowerPac HV electrophoresis system and visualized on a UV transilluminator. PCR amplification results of specimens from the Italian sexual populations were inconsistent; while initial reactions at an annealing temperature of 46 °C produced visible but multiple bands, a subsequent change to 44 °C yielded only an amplicon for a single specimen, limiting the ability to evaluate the effectiveness of the altered conditions. Even after repetition and further alteration of the annealing temperature, no amplicons were successfully generated for other specimens. All processes adhered to standard operating

procedures for molecular workflows to minimize contamination and ensure reproducible results, for example by including negative controls.

2.2. Sequence Alignment

The initial dataset comprised 190 COI sequences obtained from *Cypridopsis vidua* specimens collected across five geographic regions. Belgian specimens were sampled as part of the SPEEDY (IUAP Belspo) project between 29 May and 10 July 2013, from 33 SP-sites spanning 27 plots and 27 municipalities across Belgium, including private, natural, and city ponds. The US specimens were collected on 19 August 2012 from a garden pond with algal growth on Gardener Road, Woods Hole, Massachusetts. Ecuadorian specimens originated from Lake San Pablo in September 2018. Brazilian specimens were obtained from multiple sites in Brazil. Sequence data from additional specimens (including New Caledonia and Turkey) supplemented the dataset; most of these were downloaded from GenBank.

To root phylogenetic and population-genetic analyses, an outgroup (GenBank accession GU566803; *Macroscapha* sp.) was included, following previous usage in ostracod phylogenetics to represent a closely related but distinct lineage (Cohen & Morin, 1990). To verify that the generated sequences corresponded to *C. vidua* and to screen for potential contamination, each assembled consensus sequence was exported in FASTA format and subjected to BLASTN (v2.9.0) searches against the NCBI GenBank database (Altschul *et al.*, 1990). If a sequence matched something other than an ostracod or showed less than 90% identity to known *C. vidua* COI sequences, the sample was re-examined. Any suspicious or poor-quality sequences were re-trimmed or omitted.

All sequences were aligned using the software MAFFT v7.453 (Katoh & Standley, 2013) under default settings with the “auto” strategy, allowing the algorithm to select the most appropriate alignment method based on sequence similarity (Katoh *et al.*, 2002). The MAFFT generated alignment was exported in FASTA format and imported back into programme BioEdit (Hall T.A., 1999) for manual inspection. Each site was examined for potential misalignments, particularly in regions containing gaps or ambiguous characters. Any observed gaps were carefully inspected: if they resulted from sequencing errors, the corresponding sequences were retrimmed. After this editing process, all gaps had been removed generating a final alignment length of 492 bp. To ensure reproducibility, the alignment parameters were documented, including the substitution matrix used in MAFFT. The final edited alignment file, containing 170 *C. vidua* sequences plus the outgroup, served as basis for haplotype network construction, phylogenetic tree inference, and genetic species delimitation analyses.

2.3. Phylogenetic Analyses

The curated alignment of 170 *Cypridopsis vidua* COI sequences (492 bp), together with a single outgroup sequence (*Macroscapha* sp.; GenBank accession number GU566803), underwent analyses for statistical selection of the best-fit nucleotide substitution model with the software JModelTest v.2.0 (Darriba *et al.*, 2012) via the CIPRES Science Gateway v.3.3 (Miller *et al.*, 2010) with run times of 2.5 hours to ensure thorough parameter exploration. For JModelTest analyses, results of model selection were evaluated with both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) settings. Phylogenetic relationships were then reconstructed using both maximum likelihood (ML) and Bayesian inferences (BI). Under AIC, the GTR + G (General Time Reversible plus gamma-distributed rate heterogeneity) model was identified as optimal for ML reconstruction, while under BIC, the HKY + G (Hasegawa-Kishino-Yano plus gamma) model was recommended for Bayesian analysis.

For phylogenetic reconstructions based on the maximum likelihood method, the programme PhyML v.3.1 (Guindon & Gascuel, 2003) was used. The curated alignment in PHYLIP format was prepared using the software DnaSP v.6.0 (Rozas *et al.*, 2017). Empirical base frequencies and relative rate parameters were applied as estimated by JModelTest. The “slow and accurate” search algorithm was employed with 1,000 nonparametric bootstrap replicates to assess node support.

Bayesian analyses were carried out with the programme MrBayes v.3.2.6 (Ronquist *et al.*, 2012), using the HKY + G model. Two independent runs, each with four Metropolis-coupled Markov chains, were executed for 1,000,000 generations, sampling every 100 generations. Convergence was confirmed by average standard deviation of split frequencies dropping below 0.01 and effective sample sizes exceeding 200 for all parameters. The first 25 % of sampled trees were discarded as burn-in, and a majority-rule consensus tree was generated to obtain posterior probabilities.

Both ML and BI phylogenies were rooted with the *Macroscapha* outgroup and visualised with the software FigTree v.1.4.4 (Rambaut, 2010). Branches were annotated with bootstrap values ≥ 90 % and posterior probabilities ≥ 0.85 , and clades were colour-coded by sampling locality (South America vs. North America) to facilitate interpretation of phylogeographic structure.

2.4. Haplotype Network

To assess intraspecific genetic variation and population structure a haplotype network was constructed using the software PopArt v.1.7 (Leigh & Bryant, 2015) using the default minimum spanning network algorithm (epsilon = 0). The edited alignment (including all 170 ingroup sequences) was reformatted into interleaved Nexus format via DnaSP, ensuring that taxon names corresponded to sample identifiers and reflected geographic origin (e.g., “ECU_1,” “ECU_2” for Ecuador, “USA_1,” “USA_2” for USA).

2.5. Genetic Species delimitation

The following four complementary species delimitation methods were applied in the current study: two distance-based techniques (ABGD and ASAP) were combined with two methods based on phylogenetic frameworks (KoT and bPTP) to produce robust results for genetic species delimitation as recommended by Schön *et al.* (2018).

Two complementary distance-based delimitation tools were applied. First, the **Automatic Barcode Gap Discovery** algorithm (ABGD; Puillandre *et al.* 2012) partitioned sequences by identifying barcode gaps using a Web interface (<https://bioinfo.mnhn.fr/abi/public/abgd/abgdweb.html>). ABGD was configured with a prior intraspecific divergence (Pmin) of 0.001 and maximum divergence (Pmax) of 0.1, iterating over 10 steps and dividing data into 20 bins under the Kimura 2-parameter (K80) model. Next, **Assembling Species by Automatic Partitioning** (ASAP; Puillandre *et al.* 2021) was executed on its online server (<https://bioinfo.mnhn.fr/abi/public/asap/asapweb.html>), testing uncorrected p-distances and Kimura (K80) substitution matrices to corroborate the species partitions identified by ABGD.

The **K/θ (KoT)** method (Spöri *et al.* 2021) based on **Birky's 4 Theta/K Rule** (Birky *et al.* 2010), was applied via the KoT web server (<https://eeg-ebi.github.io/KoT/input.html>). By comparing genetic divergence (K) to genetic diversity (θ) within and between strongly supported phylogenetic sister clades, species boundaries are confirmed when the K/θ ratio surpasses 4, consistent with the genealogical species concept.

The **Bayesian Poisson Tree Processes** approach (**bPTP**; Zhang *et al.* 2013), was used to model the speciation on phylogenetic trees. It models substitution events along branches as two independent Poisson processes, one for within-species (coalescent) branching and one for between-species (speciation) branching and uses Markov chain Monte Carlo (MCMC) sampling to estimate posterior probabilities for each putative species partition. This analysis was performed on the bPTP server (<https://species.h-its.org/ptp/>) with 500,000 MCMC generations.

3. Results

A total of 199 *Cypridopsis vidua* COI sequences were analysed. The dataset comprised specimens representing populations from four major continents: Europe, North America, South America, and Oceania, thus spanning a large portion of the species' global distribution. After alignment and trimming, the final COI fragment retained for analyses was 492 bp in length. This dataset formed the basis for all subsequent phylogenetic reconstructions, haplotype network analyses, and genetic species delimitation, enabling the assessment of global patterns of genetic diversity and population structure in *C. vidua*.

It is important to note that DNA sequences from the Italian *Cypridopsis vidua* specimens could not be included in the final dataset due to technical difficulties encountered during amplification. While DNA extraction was successful, several of the samples appeared to be degenerated, and repeated PCR attempts failed to produce amplicons from the other specimens. As a result, no high-quality COI sequences could be generated in time for inclusion in the alignment or subsequent phylogenetic analyses.

3.1. Phylogenetic reconstruction

The software JModelTest identified different optimal substitution models depending on the information criterion applied. The Akaike Information Criterion (AIC) favoured the General Time Reversible model with a proportion of invariant sites and gamma-distributed rate heterogeneity (GTR + I + G), which was selected for the Maximum Likelihood phylogenetic analysis conducted in PhyML. Conversely, the Bayesian Information Criterion (BIC) supported the HKY model with gamma-distributed rate heterogeneity (HKY + G), which was used for the Bayesian phylogenetic inference. We decided to apply the AIC-selected model for the PhyML analysis and the BIC-selected model for the Bayesian analysis, allowing appropriate model choice for each method. Under the GTR + I + G model, the estimated base frequencies were A = 0.2864, C = 0.2000, G = 0.1543, and T = 0.3594, with a gamma shape parameter of 0.6630. Under the HKY + G model, base frequencies were A = 0.2880, C = 0.1896, G = 0.1438, and T = 0.3786, with a gamma shape parameter of 0.6540, indicating moderate among-site rate heterogeneity in both models.

Maximum-likelihood analysis, conducted in PhyML under the GTR + I + G substitution model, produced a tree with an overall shallow topology and limited internal resolution (see **Figure A1** in [Appendix](#)). The ingroup sequences formed a single, continuous clade without deep subdivisions. Most internal nodes within *C. vidua* exhibited very short branch lengths and showed minimal bootstrap support, resulting in a central polytomy rather than clearly delineated phylogenetic clades with strong statistical support.

At the terminal tips, several small clusters of two to four closely related specimens achieved slightly higher bootstrap values (approximately 50–60%), but these clusters did show any consistent grouping according to geographic origin or reproductive mode. For instance, pairs of specimens from Belgium and Canada occasionally grouped together with moderate support; yet, similar clusters also combined samples from Brazil and the United States, underscoring the absence of region-exclusive phylogenetic groups. Not a single clade reached robust statistical

support ($\geq 90\%$), and there were long branches that would indicate deeply divergent or cryptic mitochondrial lineages within the dataset.

Geographically, the samples were intermingled throughout the tree rather than partitioned into distinct clades. This widespread mixing of regional samples, coupled with the uniformly low bootstrap values and shallow branching. In sum, the PhyML analysis portrays *C. vidua* as a single, genetically cohesive assemblage with limited mitochondrial divergence and little phylogeographic structure.

Bayesian phylogenetic inference conducted in MrBayes produced a tree topology that was largely congruent with the results of the maximum-likelihood analyses, exhibiting a similarly shallow structure and limited internal resolution (see **Figure A2** in [Appendix](#)).

In contrast to the ML tree, the Bayesian tree displayed shorter branches lengths and generally moderate to high posterior probability (PP) values at terminal nodes, indicating some statistical support for the phylogenetic position among closely related individuals. However, support for deeper branches was notably weaker, with many internal nodes exhibiting low PP values. This pattern resulted in relatively unresolved deeper branches that form polytomies or poorly supported splits, indicating limited phylogenetic differentiation within the ingroup.

Similar to the PhyML tree, geographic intermixing was evident in the Bayesian phylogeny. Specimens from Belgium, Brazil, Canada, United States, Turkey, Mexico, and New Caledonia were distributed throughout the tree without forming well-supported, exclusive geographic clades. Several terminal branches with strong posterior support ($PP \geq 0.95$) included specimens from multiple geographic origins (see **Figure 4**), further highlighting the absence of a clear phylogeographic structure.

3.2. Haplotype Network

The haplotype network (see **Figure 3**) comprised 80 distinct haplotypes connected by multiple mutational steps, displaying a star-like structure centred around a single, high-frequency haplotype shared by specimens from Europe, North America, South America, Turkey, and New Caledonia. This dominant haplotype occupied the central position in the network and was represented by the largest circle indicating that sequences of most individuals belonged to this haplotype. Surrounding this central node were numerous different haplotypes each separated by more than 12 mutational steps from the central haplotype. In addition to these closely related variants, a subset of haplotypes was positioned markedly farther from the central haplotype, connected by longer branches comprising multiple mutational steps. These more divergent haplotypes are represented by few individuals and include specimens from New Caledonia, South America, and North America.

Geographically, there is extensive intermixing of haplotypes: samples from Europe, South and North America, Turkey, and New Caledonia are distributed throughout both the central and peripheral haplotypes. No clear, region-specific clusters formed, and different continental origins are frequently mixed within individual haplotypes (e.g., The central node includes Belgian, Canadian, Brazilian, Turkish, and New Caledonian specimens; The peripheral haplotype highlighted in magenta at the far right of **Figure 3** is restricted to Belgian and Canadian samples).

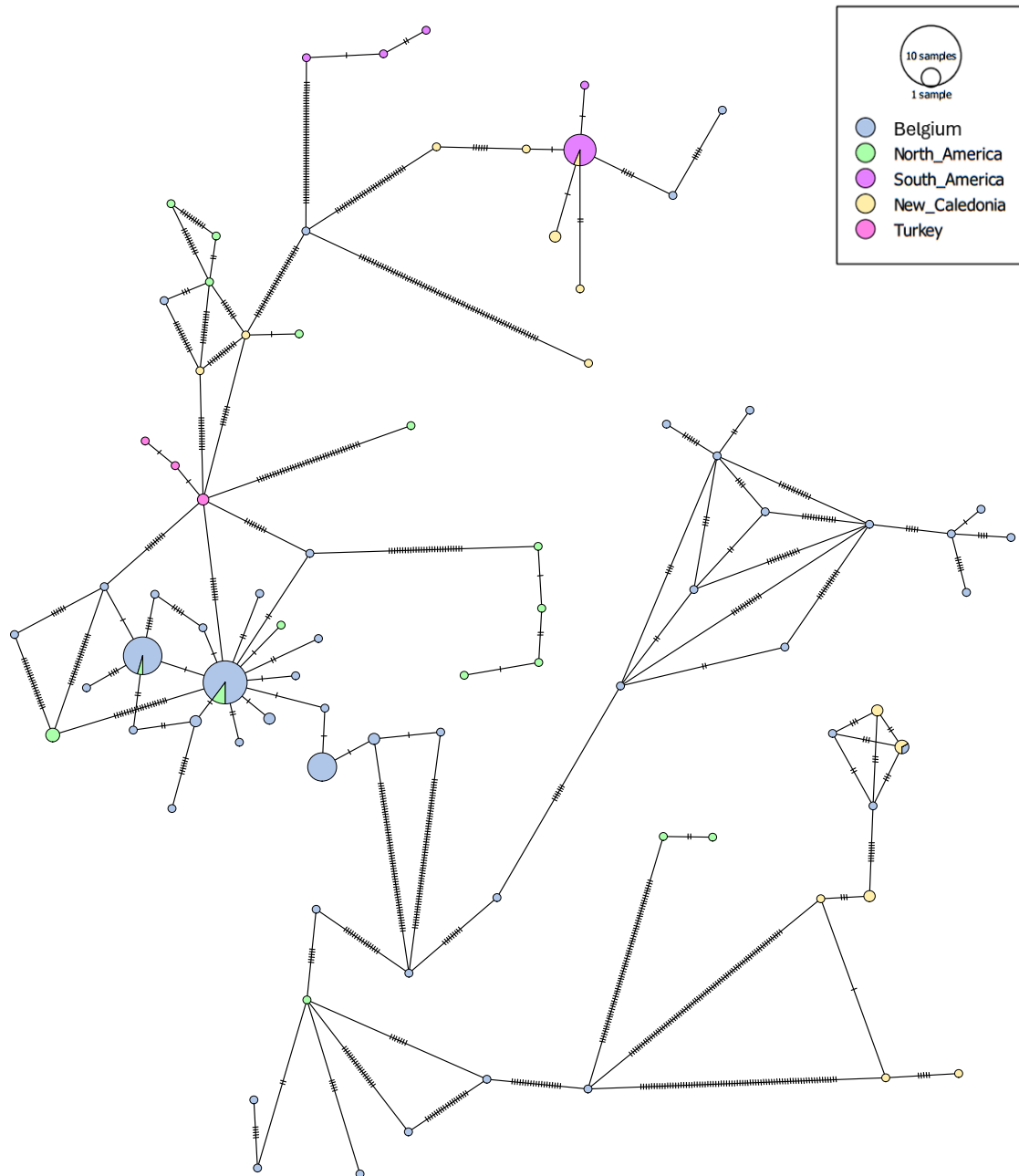


Figure 3. Haplotype network of *Cypridopsis vidua* based on COI sequences and constructed in with PopArt v.1.7 (Leigh & Bryant, 2015). Haplotypes are coloured by sampling region: light blue for Belgium; light green for North America; purple for South America; yellow for New Caledonia; and pink for Turkey. The size of each circle is proportional to the number of individuals sharing that haplotype, and each horizontal line connecting circles represents a single mutational step between haplotypes.

3.3. Genetic species delimitation

3.3.1. Automatic Barcode Gap Discovery (ABGD)

The ABGD analysis on the COI dataset, using pairwise Kimura 2-Parameter (K80) distances, generated a stable partition at a prior maximal intraspecific divergence (P) value of 0.0359. This threshold consistently resolved nine candidate genetic species (see **Figure 4**), with the grouping pattern remaining robust across recursive partitions and a narrow range of prior values (P = 0.033–0.037).

The largest group identified by ABGD (Group 1) contained 105 sequences and was widely geographically distributed, encompassing specimens from multiple continents. The remaining eight groups were smaller and typically regionally restricted; for instance, Groups 4 and 5 were composed solely of Brazilian sequences, while Group 8 contained only New Caledonian samples. The ABGD histogram of pairwise genetic distances exhibited a multimodal distribution, with a dominant peak at low divergence (<0.01) and several smaller peaks at greater distances (0.02–0.12), consistent with the chosen partitioning. The barcode gap detection remained congruent with the nine-group result, indicating stable species delimitations within the selected P threshold.

3.3.2. Assembling Species by Automatic Partitioning

The ASAP analysis was performed on the same COI data using three distance models: Kimura 2-Parameter (K80), Jukes–Cantor (JC69), and uncorrected p-distances. All three models produced broadly similar species partitions with minor ranking differences (see **Figure 4**). Using the K80 model, the optimal ASAP partition (ASAP score = 1.0) identified thirteen candidate genetic species, including one large, geographically widespread cluster analogous to ABGD Group 1, and twelve smaller groups assigned to distinct regions. Several clusters matched single geographic areas, such as a group being exclusive to North America and a cluster being restricted to New Caledonia.

The ASAP score plot revealed a series of low scores at divergence thresholds between 0.03 and 0.05 substitutions/site, suggesting that these corresponded to the most likely partitions. Higher thresholds produced fewer but larger groups, accompanied by poorer ASAP scores. Consistency across distance models confirmed that the thirteen-group result reflected genuine partitioning rather than analytical artifacts.

3.3.3. Birky's 4 Theta/K Rule (KoT)

The KoT analysis based on the K/θ species criterion ($K/\theta > 4$) identified a total of thirty-seven clusters among the COI sequences. These ranged from large, widely distributed groups to numerous small and highly divergent lineages. Many small clusters predominantly comprised Brazilian samples, reflecting pronounced regional diversification, while other clusters were composed of isolated North American, New Caledonian, or Turkish sequences.

Strong genetic separation was evident within certain clusters, particularly for some New Caledonian and North American lineages, which were distinguished by high K/θ values and substantial distances from nearest neighbour clusters. In contrast to ABGD and ASAP, KoT

separated more candidate genetic species by being sensitive to fine-scale divergences within broader mitochondrial clades.

3.3.4. Bayesian Poisson Tree Processes (bPTP)

The analysis applied to the COI phylogenetic tree identified 30 candidate genetic species, with posterior probabilities (PP) ranging from 0.53 to 1.00. The majority of these clusters received strong statistical support, with PP values equal to or exceeding 0.90, and several clusters were supported maximally (PP = 1.00), indicating high confidence in their delimitation (see **Figure 4**).

Among the clusters, one of the largest well-supported groups included multiple North American specimens, forming a coherent genetic species with full posterior support. Several clusters were exclusively composed of Belgian sequences, some of which attained maximal support, highlighting discrete genetic units within this geographic area. Similarly, Brazilian and New Caledonian specimens each formed multiple, well-supported region-specific clusters, reflecting local genetic diversity. Other clusters showed mixed geographic origins, containing specimens from various continents, which indicated some degree of genetic intermixing within these proposed genetic species.

Smaller clusters were also recovered, often representing unique or rare lineages. These small groups frequently had strong support and included single sequences or small sets of specimens from geographically limited regions. Intermediate support values were observed for certain clusters composed of Turkish specimens and combined Belgian and Canadian origins, reflecting moderate confidence in these subdivisions.

Four species delimitation methods recovered different numbers of candidate genetic species from the aligned *C.vidua* dataset. ABGD ($P = 0.0359$) generated nine clusters, comprising one large, geographically widespread group and eight regionally restricted lineages. ASAP produced thirteen clusters, including the nine ABGD groups plus four additional regional splits. bPTP identified thirty clusters, subdividing several ABGD/ASAP groups, particularly within Belgian and Brazilian samples, into multiple well-supported units. KoT recovered thirty-seven clusters, yielding the highest number of candidate species, with numerous fine-scale splits among Brazilian and Belgian sequences.

Pairwise comparisons showed that ABGD and ASAP were most similar, sharing the same widespread cluster and the majority of regional groups, differing by 4 ASAP-specific clusters. bPTP and KoT exhibited moderate concordance in the number and composition of fine-scale clusters, although KoT separated more narrowly defined lineages. The largest discrepancy was between ABGD (9 clusters) and KoT (37 clusters), reflecting the range of sensitivity across delimitation criteria.

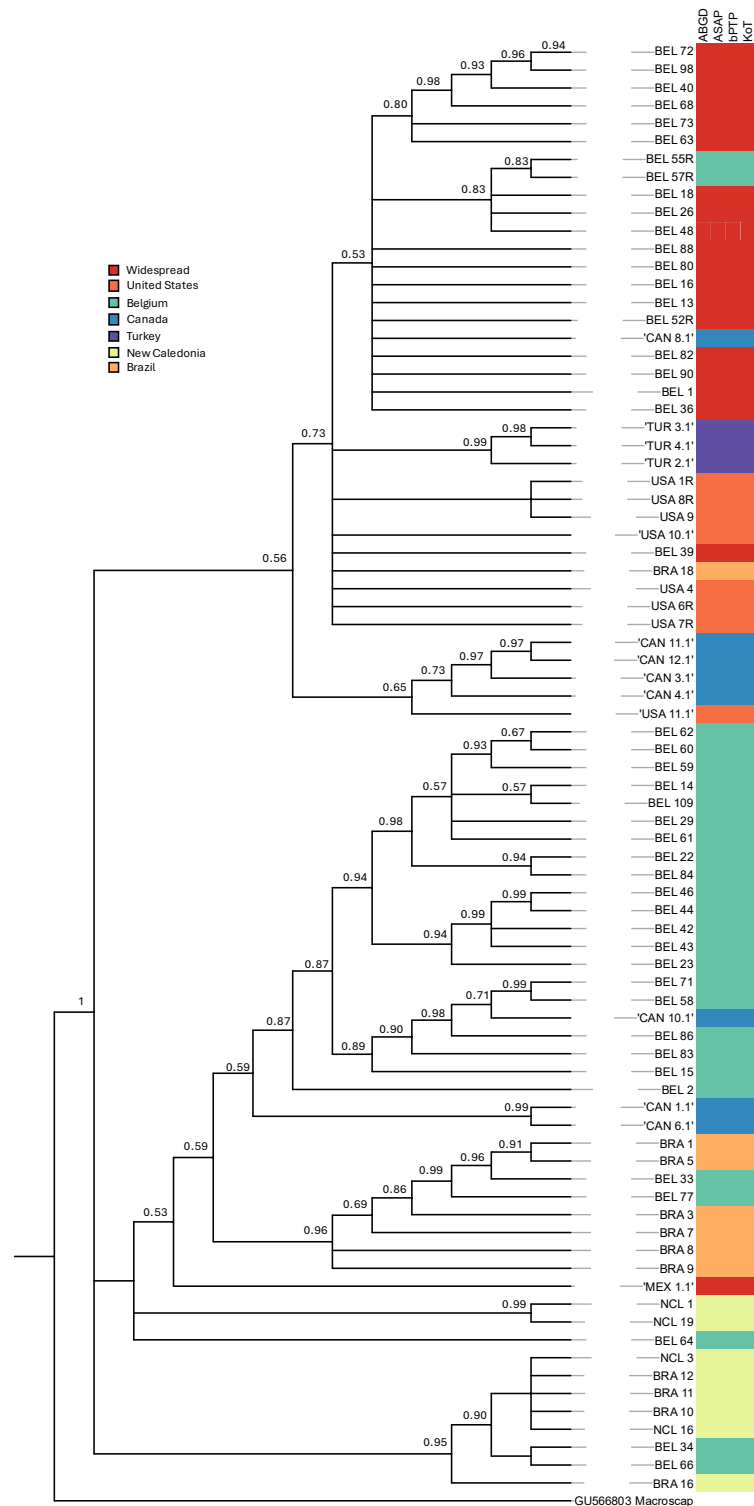


Figure 4. Bayesian COI phylogeny with posterior probabilities shown on branches. Coloured bars represent species delimitation results from ABGD, ASAP, bPTP, and KoT methods. Colours indicate geographic origins: red (widespread), while the following colours indicate exclusive geographic distribution: orange (USA), blue (Canada), green (Belgium), purple (Turkey), teal (New Caledonia), yellow (Brazil). The tree was rooted with *Macroscapha sp.* as outgroup.

4. Discussion

4.1. Amplification and Sequencing Success

The mitochondrial COI dataset compiled in the current study for *Cypridopsis vidua* draws on sequences generated by the SPEEDY project and additional COI data produced by an intern student from the master program of Oceans & Lakes and I. Schön, providing broad geographic coverage across multiple continents. Although these existing resources yielded a high proportion of quality sequences, testifying to the suitability of the COI primers and DNA-extraction protocols, the additional Italian specimens consistently failed to amplify, likely due to degraded DNA in older or poorly preserved material. Future efforts could include redesigning primers to target shorter COI fragments, optimizing extraction protocols for low-yield samples, or sourcing fresh Italian material to resolve these gaps. The assembled dataset nevertheless provides a good basis for all downstream phylogenetic reconstructions and species-delimitation analyses that investigate genetic variation, population structure, and potential cryptic speciation within *C. vidua*.

However, the geographic representation of COI sequences was constrained by data availability in GenBank, resulting in uneven sampling across continents. There are regions with few or no deposited sequences, which may bias inferences of population structure and diversity. Future work should prioritize generating COI data from these gaps to achieve a more balanced dataset.

Such a more balanced geographic sampling would strengthen the dataset's representativeness and reduce biases in downstream analyses. Yet, even with a complete and geographically balanced COI dataset, additional constraints stem from the inherent properties of the marker itself. An important methodological limitation, documented in both the literature (Rubinoff *et al.*, 2006) and reflected in this study, concerns the use of mitochondrial COI as the sole genetic marker. While COI is standard in animal barcoding, numerous studies have shown that it may be insufficient for accurate genetic species delimitation and phylogenetic reconstruction in taxa with recent divergence or slow mitochondrial evolution (Dupuis *et al.*, 2012; Rubinoff *et al.*, 2006; Kekkonen & Hebert, 2014; Solovyeva *et al.*, 2023; Yang *et al.*, 2025). In ostracods, evolutionary rates of COI can be comparatively low, limiting its resolution for deep phylogenetic branching and ancient divergence events (Schön *et al.*, 2015; Schön *et al.*, 2018). Additionally, the amplification of nuclear mitochondrial pseudogenes in some ostracod lineages can produce ambiguous or misleading COI sequences, and because mitochondrial DNA is maternally inherited, it reflects only maternal lineage history rather than the full genealogical contribution of both parents. This limitation may obscure processes such as hybridization, male-mediated gene flow, or recombination (Schön *et al.*, 2015, 2018). To obtain a more comprehensive and balanced understanding of genetic diversity in *C. vidua*, future studies should complement mitochondrial barcoding with nuclear loci (e.g., EF1 α , 28S rRNA) or genome-wide data, which are critical for detecting evolutionary phenomena like hybridization and polyploidy documented among non-marine ostracods (Kilikowska *et al.*, 2024; Schön *et al.*, 2012b; Karanovic, 2015; Adolfsson *et al.*, 2010). However, in this study, amplification of nuclear markers was not possible due to practical constraints, including the absence of suitable, widely applicable primers and the need for additional cloning of amplicons where polyploidy is suspected (Schön *et al.* 2018). This was not feasible both to limited time and

the small body size of ostracods, which restricts DNA yield and the number of PCR reactions that can be conducted for each individual. These limitations inevitably constrain the generality of the observed genetic patterns and may have led to undetected cryptic diversity, especially in geographic regions that are underrepresented in the dataset.

Looking ahead, the application of genomic-wide approaches such as RAD sequencing holds great potential for overcoming these limitations. Such methods can generate high-resolution data across thousands of loci, enabling precise detection of population structure, cryptic species, and complex evolutionary processes like polyploidy and hybridization (Christiansen *et al.*, 2021). Integrating these genomic tools will enhance both the resolution and the breadth of genetic insights in future studies of *Cypridopsis vidua* and related non-marine ostracods.

4.2. Phylogenetic Analyses

Phylogenetic reconstructions based on mitochondrial COI sequences of *C. vidua* were performed using both maximum-likelihood (PhyML) and Bayesian inferences (MrBayes). These yielded largely congruent tree topologies characterized by shallow branching and limited internal resolution. Within the *C. vidua* ingroup, branch lengths were uniformly short, and most internal nodes lacked strong statistical support, often resulting in polytomies or multifurcations rather than clearly defined subclades or hierarchical structure.

These patterns indicated minimal mitochondrial divergence across the geographically widespread samples, which spanned multiple continents including Europe, North and South America, Turkey, and New Caledonia. The extensive intermixing of specimens from diverse regions within terminal and internal branches reveals a lack of clearly isolated phylogeographic lineages or region-specific clades. Moderate bootstrap values or posterior probabilities were observed occasionally at terminal nodes (typically around 50–60%), but these generally did not correspond to distinct geographic groups or reproductive categories, reflecting weak support for deep splits within the species.

The haplotype network mirrored these patterns, revealing a star-like structure dominated by a single, highly abundant haplotype shared across continents, which was surrounded by numerous, rarer and closely related haplotypes. This pattern was completed by a small number of more divergent haplotypes separated from the others by multiple mutational steps. This pattern is characteristic of taxa exhibiting recent population expansion (Knüsel *et al.*, 2023; Avise (2000)).

Shallow phylogenetic structure and low support for internal nodes are in line with previous studies on non-marine ostracods demonstrating limited resolution when using mitochondrial COI as the sole marker (Schön *et al.*, 2012b, 2015, 2018; Hiruta *et al.*, 2016; Kilikowska *et al.*, 2023). The evolutionary rate of COI in ostracods tends to be slow relative to other crustaceans (Schön *et al.*, 2015), which can hinder the reconstruction of deeper evolutionary relationships and reduce the capacity to detect cryptic diversity. Furthermore, as mitochondrial DNA is maternally inherited, it only partly evolutionary history and does not capture the effects of hybridisation, recombination, or male-mediated gene flow, all of which have been documented in non-marine ostracods (Martens, 2023; Schön *et al.*, 2018).

Within this context, although our COI data effectively captured broad-scale genetic cohesion and haplotype sharing across the extensive range of *C. vidua*, they lacked the resolution necessary to tease apart fine-scale phylogeographic structure or deeper species-level splits. Studies of other ostracods have reached similar conclusions, emphasizing that even multi-locus Sanger markers (e.g., EF1 α , 28S rRNA, ITS) contribute only incremental insight and still fall short of fully resolving complex demographic histories and cryptic divergence. Instead, highly polymorphic markers such as microsatellites or genome-wide SNPs from RAD-seq are required to achieve the power and density needed for fine-scale phylogeography and robust species delimitation.

4.3. Reproductive Modes in *Cypridopsis vidua*

Our phylogenetic reconstructions (**Figure A1-A2**, [Appendix](#) and **Figure 4**) and haplotype network (**Figure 3**) reveal that specimens from newly identified sexual populations do not form a distinct, basal clade. Instead, haplotypes from the sexual population are scattered among those from presumed asexual populations, clustering within the globally dominant COI haplotype and its closely related variants. Neither the ML nor the Bayesian tree recovered any deep split separating the two reproductive types. This pervasive intermixing indicates the absence of a single “sexual root” lineage and contradicts a model of one ancestral transition to parthenogenesis followed by clonal diversification. Alternatively, the sexual root of the species has not yet been discovered.

In analogy with *Eucypris virens*, where mitochondrial COI and 16S data revealed 19 sexual and 25 asexual lineages scattered throughout the phylogeny (Bode *et al.*, 2010), our data supports a model of multiple independent origins of parthenogenesis in *C. vidua*. In *E. virens*, Bode *et al.* (2010) demonstrated that asexual lineages repeatedly arose from diverse sexual ancestors, rather than from a singular ancient transition, and that these clones subsequently dispersed widely. Similarly, in *C. vidua*, asexual clones likely originate locally from sexual parents and then spread across continents via desiccation-resistant eggs, capable of long-distance passive dispersal by waterbirds or wind (Vanschoenwinkel *et al.*, 2007), producing the mosaic haplotype distribution we observe. This passive long-distance dispersal transports each independently originated clone across continents, homogenizing mitochondrial variation without erasing signatures of its sexual origin. Further support comes from Adolfsson *et al.* (2010), who showed in *E. virens* that elevated ploidy and parthenogenesis have arisen multiple times from different sexual lineages. Together, these studies establish a model in which *C. vidua* most likely maintains sexual populations in localized refugia, while asexual lineages repeatedly originate and when hybridization occurs between sexual and asexual forms, can give rise to polyploid offspring, rather than undergoing facultative mode shifts within a single lineage.

To resolve the timing, frequency, and directionality of these transitions, high-resolution multilocus data (e.g., RAD-seq SNPs) will be essential. Such genome-wide approaches can distinguish recent gene flow from deeper phylogenetic splits, clarify the evolutionary pathways to parthenogenesis, and illuminate how *C. vidua*’s reproductive-mode dynamics have underpinned its global success. However, to achieve these goals, molecular analyses must expand beyond the single sexual population included to date; additional sexual populations, particularly from regions exhibiting geographic parthenogenesis in related species such as *E. virens*, should be incorporated.

4.4. Genetic Species Delimitation and Cryptic Diversity

Applying multiple molecular species delimitation methods (ABGD, ASAP, KoT, and bPTP) to the *C. vidua* COI dataset revealed both congruencies and discrepancies in inferred species boundaries between the different methods. The total number of candidate genetic species varied considerably depending on the method: ABGD identified nine, ASAP thirteen, bPTP thirty, and KoT thirty-seven, respectively. These differences highlight well-known challenges in defining genetic species boundaries and emphasize the importance of an integrative approach combining molecular, morphological, and ecological data (Karanovic, 2015; Pfenninger & Schwenk, 2007).

The most congruent results were obtained from ABGD and ASAP, which both retrieved a single widely distributed core lineage encompassing most samples from all sampled continents, complemented by several smaller, geographically restricted clusters (e.g., in Brazil and New Caledonia). Similar methodological concordance between gap-detection or distance-based approaches has been reported in other non-marine ostracods, such as *Heterocypris salina*, where Kilikowska et al. (2024) found only small differences in the number of candidate species depending on the program used.

KoT and bPTP, both of which are more sensitive to fine-scale divergence, produced substantially higher number of genetic species for *C. vidua*, often by splitting the broader clusters delimited by ABGD and ASAP into multiple, smaller operational taxonomic units (OTUs). In some cases, the genetic distances between these newly delimited OTUs were relatively low (e.g., < 3 % p-distance), suggesting that they may reflect population-level structure or high intraspecific variation rather than fully reproductively isolated species. Such sensitivity to shallow divergences has been noted in other taxa when applying coalescent-based or divergence-ratio methods (Birky et al., 2010; Schön et al., 2018) and may lead to overestimation of genetic species diversity when only mitochondrial markers are used (Schön et al., 2015; Rubinoff et al., 2006).

Importantly, despite the variation in absolute numbers, all methods consistently recovered the same major widespread genetic species, which encompasses specimens from Belgium, North and South America, Turkey, and New Caledonia. Similar patterns, namely a dominant, globally distributed genetic lineage, have been observed in *H. salina* (Kilikowska et al., 2024) and in *E. virens* (Bode et al., 2010). The persistence of this dominant genetic unit across different delimitation approaches, combined with the star-like haplotype network and shallow phylogenetic structure, supports the interpretation that *C. vidua* represents a genetically cohesive species with limited mitochondrial differentiation across its global distribution.

The smaller clusters identified by more sensitive methods showed considerable geographic mixing rather than clear regional exclusivity. For instance, clusters often contained Belgian specimens alongside North American or Brazilian samples, indicating ongoing genetic connectivity rather than geographic isolation. Such intercontinental gene flow is readily facilitated by the remarkable dispersal abilities of ostracods: their desiccation-resistant resting eggs can adhere to waterbird feathers or feet and survive long-distance transport by birds or wind (McKenzie & Moroni, 1986; Vanschoenwinkel et al., 2007), and human-mediated vectors, including ballast water, aquaculture equipment, and ornamental plants, provide additional pathways for passive translocation. This

pattern contrasts with studies of endemic lake ostracods, where clear geographic structuring and region-specific lineages are commonly observed (Schön *et al.*, 2012b; Karanovic, 2015).

The genetic homogeneity observed in *C. vidua* across continents, evidenced by the dominant widespread haplotype and extensive geographic mixing in phylogenetic analyses, suggests either recent demographic expansion from a common source or high levels of ongoing gene flow facilitated by dispersal mechanisms. Such patterns have been documented in other widely distributed ostracods where mitochondrial cohesion persists despite broad geographic ranges (Schön *et al.*, 2015, 2018).

The reliance on mitochondrial COI alone may limit the detection of more recent evolutionary processes or fine-scale population structure that could be revealed by nuclear markers. However, given the extensive genetic mixing and shallow divergences observed, it is likely that nuclear data would support the primary conclusion of genetic cohesion rather than reveal extensive cryptic diversity in this species (Schön *et al.*, 2018; Rubinoff *et al.*, 2006).

4.5. Conclusion

In addressing the geographic distribution of genetic variation, the extent of cryptic diversity, and the influence of sexual reproduction on the evolutionary history of *Cypridopsis vidua*, this thesis combines mitochondrial COI sequence analysis from Europe, the Americas, and New Caledonia with phylogenetic reconstruction and four complementary delimitation methods (ABGD, ASAP, bPTP, KoT) of genetic species.

A unique contribution of this thesis is the first comprehensive molecular and morphological documentation of a sexual population of *C. vidua*. While the species has long been considered predominantly parthenogenetic (Smith *et al.* 2015; Bickford *et al.* 2007), the discovery and analysis of sexual individuals, supported by both molecular and morphological data, distinguishes this work from previous studies. The presence of sexually reproducing populations, which are rarely documented in this species, offers critical new insight into the evolutionary and genetic dynamics shaping *C. vidua*.

First, *C. vidua* exhibits remarkably shallow mitochondrial divergence and extensive haplotype sharing across all sampled regions. Both maximum-likelihood and Bayesian phylogenies are characterised by short internal branches and low node support, and the star-like haplotype network is dominated by a single, globally distributed haplotype. These patterns demonstrate that *C. vidua* functions as a single, genetically cohesive lineage rather than a suite of geographically isolated populations.

Next, while a dominant, globally distributed lineage emerges across all analyses, both ABGD and ASAP also identify several additional genetic clusters within *C. vidua* at the genetic species level. These clusters, characterized by low COI p-distances and lacking strict geographic boundaries, reveal potentially cryptic diversity within the nominal species. Rather than dismissing them as spurious partitions, our results suggest the existence of multiple genetic lineages that warrant morphological and ecological validation.

Moreover, the pervasive mitochondrial cohesion observed in *C. vidua* reflects its status as a species comprising both sexual and asexual populations rather than being strictly parthenogenetic. Periodic recombination among known sexual lineages (Rossetti *et al.*, 2024) serves to homogenize mitochondrial variation and prevent the deep divergences typical of obligate parthenogens. In strictly parthenogenetic ostracods, long-term clonal isolation produces multiple, deeply divergent mitochondrial lineages. Comparative studies of mixed-mode reproductive ostracods confirm that even infrequent sexual events can maintain genetic connectivity across vast geographic scales (Bode *et al.* 2010; Martens *et al.* 2013; Schön *et al.* 2015).

Finally, this work highlights the limitations of relying solely on a single mitochondrial marker. The slow evolution and maternal inheritance of COI constrain the resolution of recent divergence events and finer scale phylogeographic structure. To build upon these findings, future research must integrate nuclear markers (e.g., EF1 α , 28S rRNA) or genome-wide approaches, broaden geographic sampling in underrepresented regions and with additional sexual populations, and employ an integrative taxonomic framework combining molecular, morphological, and ecological data.

In conclusion, mitochondrial evidence supports viewing *Cypridopsis vidua* as a genetically cohesive species that includes both sexual and asexual populations, rather than being entirely parthenogenetic. However, the COI data also uncover multiple candidate genetic species within the nominal species, indicating potential cryptic diversity that must be rigorously tested. A multi-locus, integrative framework, combining nuclear markers, genome-wide data, morphology, and ecology, will be essential to validate these lineages, refine species boundaries, and illuminate how shifts between reproductive modes have driven the adaptability and evolutionary history of this widespread ostracod.

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5.1. Software and Tools

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6. Appendix

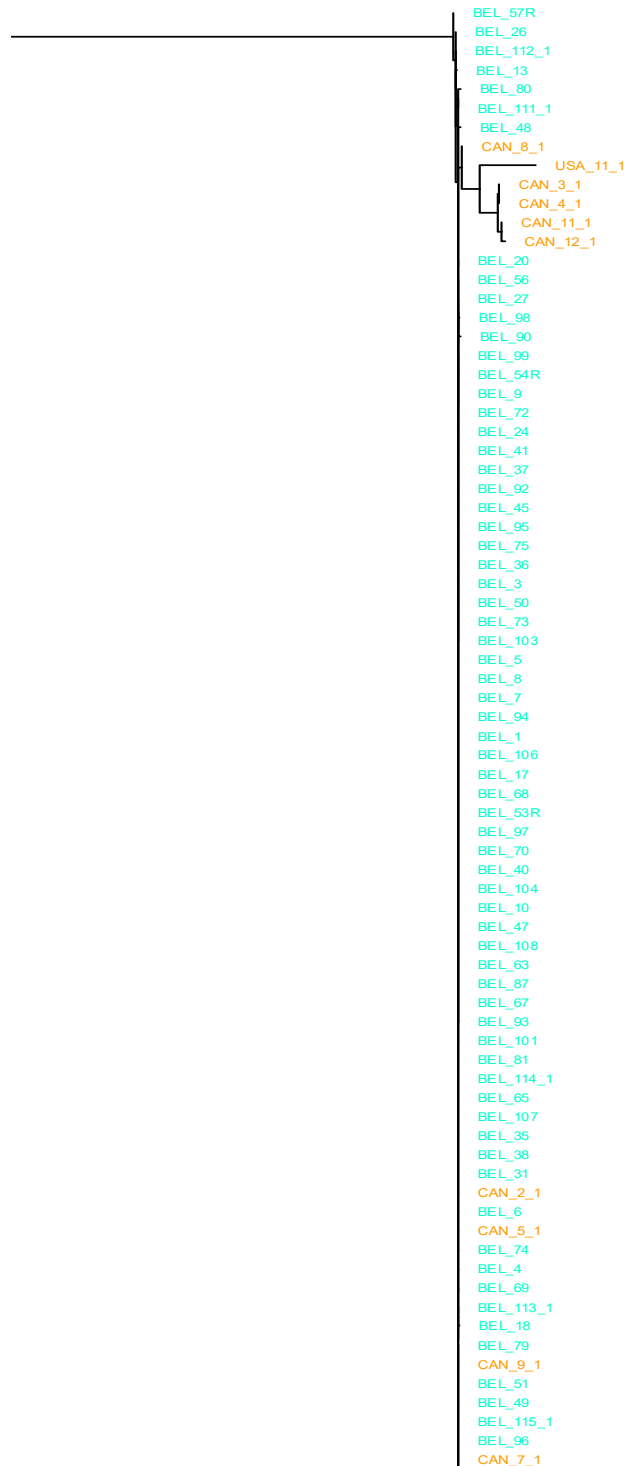
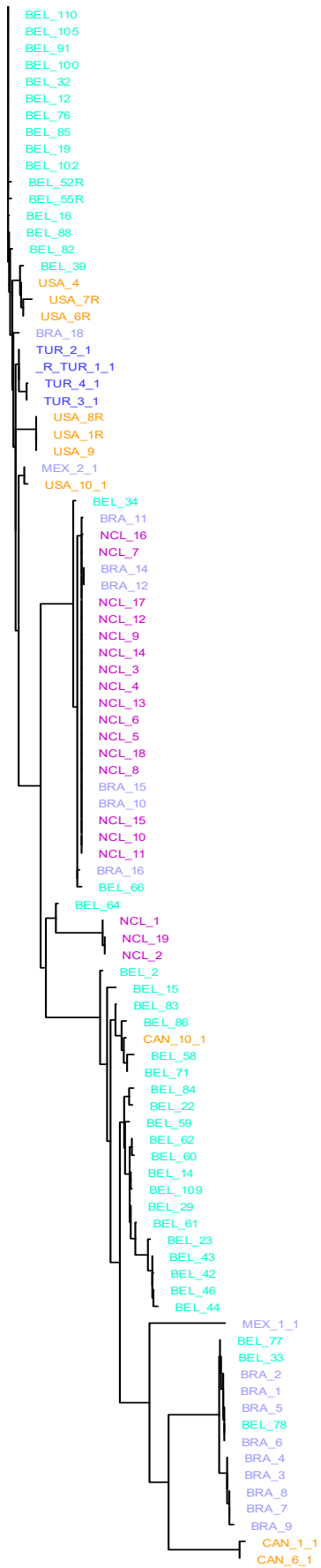


Figure A1. Maximum-likelihood COI phylogeny of *Cypridopsis vidua* constructed in PhyML v.3.1 (Guindon & Gascuel, 2003) under the GTR+I+G substitution model selected by AIC. Node support values represent bootstrap percentages from 1,000 replicates. The outgroup is *Macroscapha sp.* Branch lengths are proportional to the number of substitutions per site. *Cypridopsis vidua* COI sequences colored by sampling location. Light blue corresponds to samples from Belgium; orange represents samples from the USA and Canada; purple indicates Brazil and Mexico; magenta denotes New Caledonia; and dark blue marks samples from Turkey. (Figure continues in the next page)



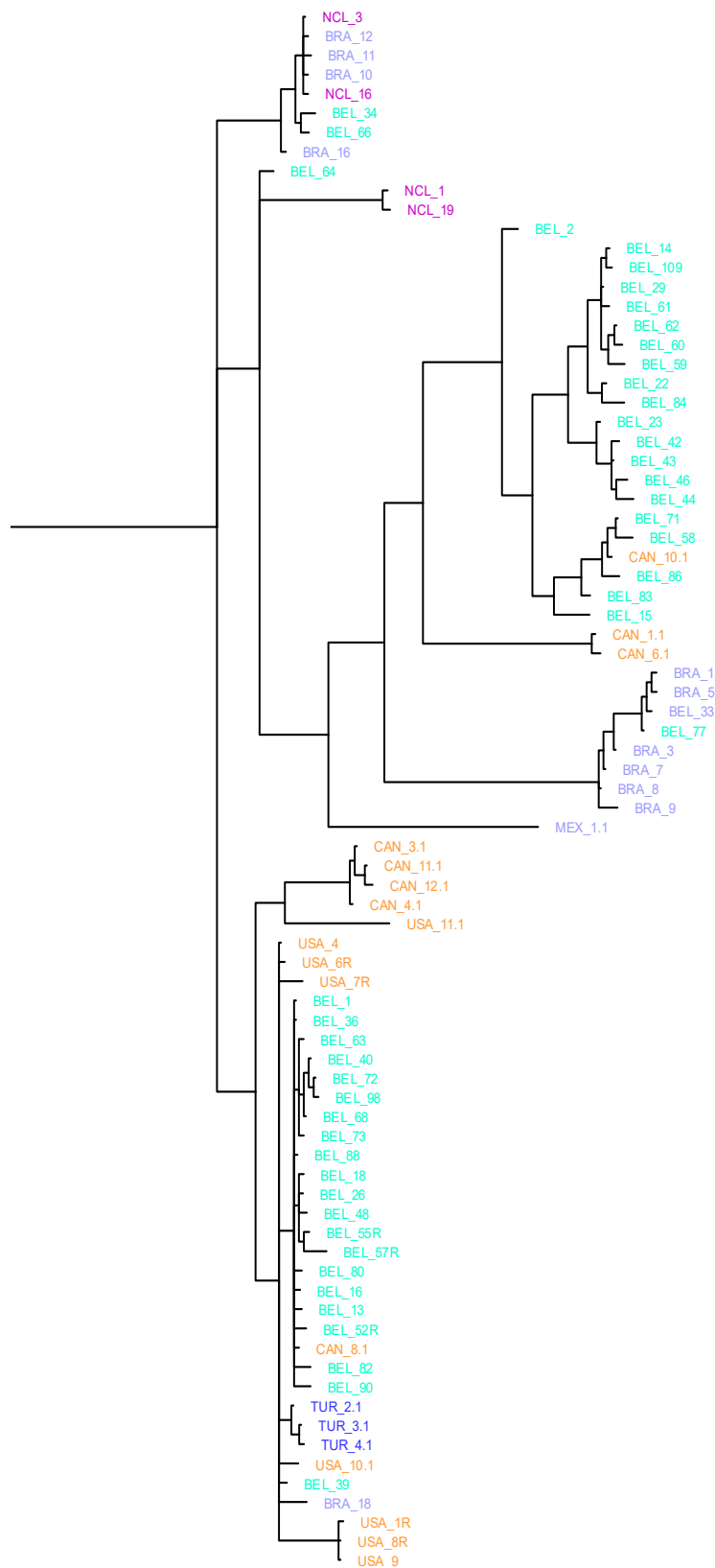


Figure A2. Bayesian COI phylogeny of *Cypridopsis vidua* constructed in MrBayes v.3.2.6 (Ronquist *et al.*, 2012) under the HKY + G substitution model selected by BIC. Node support values represent posterior probabilities from MCMC analysis. The outgroup is *Macroscapha sp.* Branch lengths are proportional to the number of substitutions per site. *Cypridopsis vidua* COI sequences are colored by sampling location. Light blue corresponds to samples from Belgium; orange represents samples from the USA and Canada; purple indicates Brazil and Mexico; magenta denotes New Caledonia; and dark blue marks samples from Turkey.